A) **ANCHOR SCENES (3–6)**

SCENE #1: Harvard to Cambridge – Setting the Stage for Hemoglobin
- **Summary:** A young Wyman sails to Europe in 1924 with classmate John T. Edsall to immerse himself in the new science of biochemistry at Cambridge. Amid a bustling lab in Sir Frederick "Hoppy" Hopkins' department, Wyman witnesses foundational discoveries about hemoglobin that spark his lifelong interest in cooperative binding[1][2].
- **Date/Place:** June 1924 – Department of Biochemistry, Cambridge University (UK).
- **People:** Jeffries Wyman (age 23); John T. Edsall (Harvard friend); Sir Frederick Gowland Hopkins (Nobel biochemist and department head); J.B.S. Haldane (visiting lecturer); Gilbert Adair (researcher on hemoglobin).
- **Concrete details:** Wyman and Edsall arrived on a slow steamer **packed with 700 cattle and 200 human passengers** – a vivid if malodorous voyage for two idealistic grads[1]. In Cambridge's new biochemistry lab, they found **J.B.S. Haldane reading Enzymes** (literally writing the book on enzyme kinetics) and teaching courses[3]. During that very term, **G.S. Adair measured hemoglobin's molecular weight** and derived the first equation for oxygen binding, overturning old assumptions[2]. Wyman soaked up these breakthroughs. After one term, he was so intrigued by another project – A.V. Hill's studies on muscle – that he shifted to University College London to pursue a Ph.D. on muscle thermodynamics[4][5].
- **Why it matters:** This scene cemented Wyman's scientific direction. He directly observed the **birth of protein cooperativity theory** (Adair's work) and gained grounding in **physical chemistry** under luminaries like Hopkins and Hill. The experience forged his **thermodynamic approach to biology**, which later proved crucial in explaining allostery[6][7]. It also began his 75-year friendship with Edsall, a partnership that would yield major publications[8][9].
- **Evidence quality:** Primary/secondary blend – Wyman's NAS memoir (by Alberty & Di Cera) draws on Edsall's first-hand recollections and Wyman's own accounts[10][2]. Strong contemporaneous context from published work (Adair 1925) corroborates the scientific happenings.
- **Myth risk:** Low – Details come from primary sources (Edsall/Wyman writings). The cattle-boat anecdote might seem colorful, but it's documented in Wyman's memoir and Edsall's writings[1].
- **Sources:** Alberty & Di Cera NAS Biographical Memoir (2003)[1][2]; Simoni *et al.*, JBC Classic (2002)[11][12].

SCENE #2: Science Detoured by War – The Emperor's Biologist
- **Summary:** In mid-1950, Wyman leaves Harvard to lecture in Korea, only to

have history intervene. The outbreak of the Korean War strands him in occupied Japan, where within days he is summoned to meet Emperor Hirohito – not as a diplomat, but as a fellow biologist[13][14].
- **Date/Place:** June 1950 – Tokyo and the Imperial Palace's biological gardens, Japan (U.S.-occupied Japan, start of Korean War).
- **People:** Jeffries Wyman (Harvard professor turned accidental envoy); General Douglas MacArthur's staff; Emperor Hirohito of Japan (an amateur marine biologist).
- **Concrete details:** Wyman had intended to give scientific lectures in Korea, but **war erupted as his ship reached Japan**, halting all travel[13]. Making the best of fate, Wyman spent six months in postwar Japan under Gen. MacArthur's administration[13]. In an almost surreal turn, Emperor Hirohito – infamous to the world as Japan's wartime leader – **invited Wyman to his summer palace to talk biology**[15]. Hirohito proudly showed his collection of **marine invertebrates**, engaging with Wyman scientist-to-scientist. Wyman wrote delighted letters about this visit to his children, describing the Emperor's avid interest in collecting animals[15][16]. (Those letters were later preserved by Wyman's daughter in a book[16].)
- **Why it matters:** This scene reveals Wyman's **adaptability and international ethos**. It marks a turning point where he transitions from a Harvard academic into a **global science ambassador**. The personal encounter with Hirohito humanized science's role in diplomacy and likely influenced Wyman's decision to work abroad (he would soon become a UNESCO science envoy)[17][18]. For his narrative, it shows science as a unifying language – even bridging two former enemy nations.
- **Evidence quality:** Primary – documented by Wyman's own letters (cited in his daughter's compilation) and referenced in his NAS memoir[15]. Secondary sources (NAS memoir) directly quote those letters' existence[19]. Very credible.
- **Myth risk:** Low – While the scenario sounds "too good" (a random American scientist advising an Emperor), it's well-documented by Wyman's correspondence[19]. No embellishment in sources beyond factual description.
- **Sources:** Alberty & Di Cera (NAS Memoir 2003)[13][19]; A.C. Wyman, *Jeffries Wyman: Letters from Japan 1950* (2000) – cited via NAS memoir.

SCENE #3: Adventure in the Hindu Kush – A Yak and a Narrow Escape
- **Summary:** In the mid-1950s, during his tenure as a roving UNESCO science officer, Wyman's penchant for adventure takes him **off the map in Afghanistan**. A casual vacation hike turns into a geopolitical incident when local authorities notice the American professor wandering in forbidden territory, forcing an unusual alpine evacuation[20].
- **Date/Place:** circa 1956 – Tribal areas at the Pakistan–Afghanistan border

(during Wyman's UNESCO service based in Cairo).
- **People:** Jeffries Wyman; local Pashtun chieftain and armed tribesmen; an assigned bodyguard.
- **Concrete details: "Jeffries was a great walker and quite adventurous,"** colleagues noted dryly[21]. On one trek in western Pakistan, Wyman unknowingly **crossed into a closed part of Afghanistan**, a region off-limits to foreigners[20]. The local chieftain was alarmed: an unbidden outsider, even a scientist, could be a spy. He ordered that Wyman be turned back immediately. Thus Wyman found himself under guard and instructed to leave – **escorted on yak-back by an armed bodyguard to the safety of the Pakistani border**[20]. One can picture the gray-haired professor in dusty hiking boots, bemused but compliant atop a lumbering yak as rifle-toting tribesmen ensure he exits their mountain pass.
- **Why it matters:** This scene, equal parts **comic and harrowing**, illuminates Wyman's **fearless curiosity and personal intrepidness**. It's more than an anecdote: it exemplifies how Wyman embraced experiences outside the lab. That worldliness informed his science – he was open to unexpected detours, whether in travel or theory. Narratively, it provides a dramatic human backdrop just before Wyman re-focuses on research. Indeed, shortly after, he settled in Rome to resume full-time scientific work[22]. The contrast makes his later theoretical breakthroughs even more striking: the adventurer-scientist who could ride a yak one year and derive elegant equations the next.
- **Evidence quality:** Primary/secondary – recounted in Wyman's NAS memoir, which likely drew on Wyman's own stories or letters and Edsall's biography[20]. The source is credible (National Academy biographers with access to Wyman's personal accounts).
- **Myth risk:** Medium – This story is less widely known (so not oft-repeated), but it's so picturesque one might suspect embellishment. However, it's explicitly recorded in a reputable source[20]. We flag it only because it lacks a second independent source. It has the ring of truth (Wyman truly *was* known for offbeat excursions[21]), but we rely on his memoir's accuracy.
- **Sources:** Alberty & Di Cera (NAS Memoir 2003)[23]; Edsall, "Philosopher and Adventurer" (Biophys. Chem. 1990) – likely source of anecdote, via memoir[24].

SCENE #4: The Allostery "Eureka" – Paris Seminar with Monod
- **Summary:** In 1964, Wyman delivers a seminar in Jacques Monod's lab in Paris that sparks a **revolutionary idea in biochemistry**. Over conversations and chalkboard sketches, Wyman, Monod, and Jean-Pierre Changeux formulate the concerted **Monod–Wyman–Changeux (MWC) model** of allosteric proteins[25][26]. This culminates in a classic 1965 paper – a eureka moment where decades of Wyman's thermodynamic thinking find a unifying expression.

- **Date/Place:** Autumn 1964 – Institut Pasteur, Paris, France (Jacques Monod's laboratory).
- **People:** Jeffries Wyman (visiting scientist from Rome); Jacques Monod (French biologist, future Nobel laureate); Jean-Pierre Changeux (Monod's young colleague); Francois Jacob (Monod's collaborator, co-Nobelist, present in milieu).
- **Concrete details:** Wyman was by then a veteran biophysicist (age 63) on a European sabbatical from Rome. Monod, fresh from Nobel-winning work on gene regulation, had coined the term "**allostery**" in 1961 but lacked a satisfying model[27]. Wyman had deep knowledge of hemoglobin's cooperative binding but approached it in thermodynamic terms. At Monod's invitation, Wyman **gave a seminar on hemoglobin conformational changes and ligand binding**[28]. One can imagine Wyman at the blackboard, sketching sigmoid curves and binding polynomials for Monod's team. The discussion lit a spark: together Monod, Wyman, and Changeux hypothesized that a protein like hemoglobin **must exist in two distinct states (tense T and relaxed R)** in equilibrium[26]. **All subunits switch state concertedly**, explaining cooperative binding in a simple, elegant way[26]. Over ensuing weeks, they fleshed out this idea into a manuscript. Monod later recounted how intensely they debated each assumption (Wyman's thermodynamic rigor balancing Monod's intuition)[29]. In 1965, their paper "On the Nature of Allosteric Transitions – a Plausible Model" was published after initial rejections[30]. It immediately provided a theoretical framework for enzymes and proteins that showed sigmoidal kinetics[31][32].
- **Why it matters:** This scene is **Wyman's scientific triumph**. The MWC model became a cornerstone of biochemistry, taught in every textbook as the concerted model of allostery[31]. For Wyman personally, it vindicated his long-held belief that thermodynamics could explain biological regulation. It also exemplified **collaboration across generations** – a senior American theorist and younger French experimentalists converging to solve a problem. The "eureka" of the Paris seminar not only transformed our understanding of hemoglobin and enzymes, but also cemented Wyman's legacy in the field. In narrative terms, this is the high point where the threads of Wyman's story (physics, proteins, international exchange) weave together into a tangible discovery.
- **Evidence quality:** Primary – Wyman's own recollections (in a 1979 tribute) and Monod's accounts document the conception of the model[28][29]. The NAS memoir provides a detailed description of the seminar and its outcome, drawn from those first-hand sources[28][26]. The published 1965 paper stands as concrete evidence of what they devised[33].
- **Myth risk:** Low – This discovery story is well documented by the participants. One potential embellishment could be the notion of an instantaneous "flash of insight." In reality the model was refined over months. But all sources agree the

pivotal **seed was planted during Wyman's 1964 visit**[28]. We have Monod and Changeux themselves crediting Wyman's role, so there's no "too perfect" legend beyond what the principals described.
- **Sources:** Alberty & Di Cera (NAS Memoir 2003)[28][26]; Monod, Wyman & Changeux, *J. Mol. Biol.* (1965) – via Wikipedia summary[31][32]; Changeux, *Origins of Molecular Biology: Tribute to Monod* (1979) – cited in NAS memoir[29].

SCENE #5: Founding EMBO – The Diplomat Scientist
- **Summary:** In 1964–1965, Wyman helps launch the **European Molecular Biology Organization (EMBO)**, leveraging his trans-Atlantic experience to knit together war-torn Europe's biologists. In a decisive committee meeting, he is appointed **EMBO's first Secretary General**, orchestrating a new era of international scientific cooperation[34][35].
- **Date/Place:** Early 1964 (planning meetings in Ravello, Italy) through 1965 (EMBO formally established; secretariat in Geneva/Brussels).
- **People:** Jeffries Wyman; John Kendrew (UK structural biologist); Sidney Brenner, François Jacob, Max Perutz (leading scientists at the Ravello meeting); European visionaries like Giuseppe "Buzz" Buzzati-Traverso and Raymond Appleyard; national science officials.
- **Concrete details:** Still based in Rome in the 60s, Wyman became one of the few Americans involved in European science policy[36]. At a 1963 conference in Ravello, a small group debated creating a pan-European biology network. There were two camps: one backing a big central lab (led by Kendrew, Brenner) and another favoring a looser network of researchers (Waddington's idea)[37][38]. Wyman, with his UNESCO background, acted as a **consensus-builder**. In 1964, as plans crystallized, the committee tapped Wyman – "formerly of Harvard, working on hemoglobin in Rome" – to be Secretary General of EMBO[34]. He was one of five members representing Italy on the first executive committee[34]. Colleagues recall Wyman as a steady hand: **"a tower of strength" who kept even Nobel laureates like Perutz on task** when scientific egos might have derailed progress[39]. He helped draft EMBO's constitution and recruit its initial 200 members[40][41]. Wyman served as Sec-Gen from 1965 to 1969, guiding decisions like where to situate the new European Molecular Biology Laboratory (EMBL)[42][43]. Notably, he mediated between scientists and government funders (Euratom, etc.), using skills honed during his UNESCO days[44][39].
- **Why it matters:** This scene underscores Wyman's role as a **scientific statesman**. It might lack the flash of a singular discovery, but narratively it provides resolution: Wyman giving back to the global community that he loved. The organization he helped create (EMBO) went on to profoundly shape molecular biology – a legacy arguably as enduring as his allostery model. For Wyman, EMBO was a capstone where his **ideals of collaboration and his**

breadth of connections paid off. It also provides a satisfying "full circle": the young man who studied in Europe returns in later life to unify European science.
- **Evidence quality:** Primary/secondary – EMBO's 40th and 50th anniversary histories detail Wyman's contributions, including quotes from colleagues like Brenner and Appleyard that highlight Wyman's character[39][45]. The NAS memoir also notes his position as a founder and officer of EMBO[18]. These are well-sourced from organizational records and letters (e.g. Wyman's correspondence with Perutz in 1964[44]).
- **Myth risk:** Low – There's little mythology here; Wyman's EMBO role is sometimes underplayed rather than exaggerated. One could misattribute EMBO's founding solely to bigger names (Kendrew, Brenner), but records clearly show Wyman's key role[36][39]. We have multiple corroborating documents.
- **Sources:** EMBO 50th Anniversary Book (2014)[36][39]; Alberty & Di Cera (2003)[18]; Brenner oral history via EMBO history[39].

B) **DIRECT QUOTES (3–7)**

- *"In the course of reading over the other day, at a window by the sea, the page proof of an article on linkage I was suddenly struck by the realization that in all the years I had been thinking about the matter I had consistently failed to recognize one significant general concept… This is the concept of what may be called the binding potential."*[46][47]
  Context/Date: From Wyman's own introduction to his paper *"The binding potential, a neglected linkage concept,"* published 1965 in *Journal of Molecular Biology*. Wyman poetically describes the **"aha moment"** when he conceived the binding potential idea while reviewing galley proofs by the seaside.
  Source: Wyman 1965, quoted in NAS Biographical Memoir (Alberty & Di Cera 2003)[46][47].
- *"Thermodynamic potentials 'are not accessible to measurement but are known only in terms of their changes.'"*[48]
  Context/Date: Wyman's explanatory aside in the same 1965 "binding potential" paper, clarifying a point of physical chemistry for biologists. He emphasizes a fundamental principle: you can never measure an absolute energy potential, only changes in it (mirroring how one measures changes in free energy, not absolute values).
  Source: Wyman (1965), as cited by Alberty & Di Cera[49].
- *"Every ligand may be expected to exert some influence on every other; in other words, that, at the highest level of approximation, the ligands all form a single linkage group."*[50]
  Context/Date: Wyman (1965) in the binding potential paper, stating the

principle of linked functions in clear terms. Here he generalizes that in a multi-site binding protein, all binding sites and effectors are thermodynamically linked (a conceptual foundation for allostery). It's Wyman's succinct description of complete coupling among ligands.
Source: Wyman 1965, via Alberty & Di Cera (NAS memoir)[50].

- *"Why is it, from a molecular point of view, that the different conformations should have different ligand affinities, and why is it that the uptake of ligand should lead to a conformation change?"*[51]
  Context/Date: Wyman, in the concluding remarks of his comprehensive 1968 review on regulation in macromolecules[52]. Here he articulates the core questions driving the study of allostery – essentially asking **why proteins have different states (affinities) and why ligand binding switches those states**. This quote exposes Wyman's inquisitive mindset, framing the puzzle that the MWC model and subsequent theories aim to solve.
  Source: Wyman 1968 (Advances in Protein Chemistry review), quoted in NAS Biographical Memoir[51].
- *"[These] problems of structure and function lie in the Biophysics and Biology of tomorrow."*[53]
  Context/Date: The very next lines of Wyman's 1968 review[53], as he wraps up his discussion of linked functions. It's both a humble admission that unanswered questions remain and a forward-looking prediction. Wyman acknowledges that fully understanding allostery and conformational change will be the work of the next generation ("tomorrow"). This quote shows Wyman's *future-oriented vision* and modesty about what had been achieved so far.
  Source: Wyman 1968, concluding sentence, via NAS memoir[53].
- *"Jeffries Wyman was a great walker and was quite adventurous."*[21]
  Context/Date: Comment in Wyman's NAS biographical memoir (2003) describing his personality during the mid-1950s (his UNESCO period). Likely attributed to John Edsall or another colleague, this remark introduces the anecdote of Wyman's Afghanistan escapade. It reads almost like an understatement, given the ensuing yak story, but serves as a character quote showcasing how those who knew him viewed his temperament for adventure.
  Source: Alberty & Di Cera (quoting Edsall's 1990 memoir or similar)[21].
- *"His old friend Jeffries Wyman recalled that in Oppenheimer's later years he was 'terribly upset by his whole connection with the bomb.' … Wyman got the impression that his friend 'was almost reveling in his feeling of guilt about the bomb.'"*[54][55]

Context/Date: Wyman's reflections in a 1975 oral history interview about J. Robert Oppenheimer. Wyman knew Oppenheimer from their student days and met again in the 1950s. Here, speaking to historian Charles Weiner, Wyman describes Oppenheimer's conflicted emotional state regarding the Manhattan Project in retrospect – a mix of genuine anguish and self-indulgence in guilt[54][55]. This quote illuminates Wyman's perceptiveness about a friend's psyche, not to mention placing him quietly in the drama of atomic ethics.
Source: Jeffries Wyman interview by C. Weiner (28 May 1975), as cited in Hijiya's *"The Gita of J. Robert Oppenheimer"* (Proc. Amer. Phil. Soc. 2000)[54][55].

C) **HUMAN REALITY DETAILS (8–15 bullets)**

- **New England lineage:** Wyman was the **third Jeffries Wyman** in his family; his grandfather (Jeffries Wyman, 1814–1874) had been a famed Harvard anatomist and a founding member of the NAS[56]. His father worked for Bell Telephone. He grew up in Massachusetts with science and Harvard in his blood. — *Source: NAS Memoir*[56]
- **Philosopher-scientist:** At Harvard College (entered 1919), Wyman **double-majored in philosophy and biology**, graduating *summa cum laude* in philosophy and with high honors in biology[57]. He approached science with a philosophical mindset, an unusual blend that gave him a broad intellectual outlook. — *Source: NAS Memoir*[57]
- **Thermodynamics early on:** As an undergrad, he took physicist P.W. Bridgman's renowned course in advanced thermodynamics, developing a "remarkable aptitude" for it[6]. This rigorous training in physical chemistry would later underpin his theoretical models in biochemistry. — *Source: NAS Memoir*[6]
- **Lifelong friendship:** Wyman's college friendship with **John Tileston Edsall** lasted over 75 years[58]. The two bonded as Harvard juniors and remained close collaborators. (They even co-authored the classic textbook *Biophysical Chemistry* in 1958 and were still corresponding in their 90s.) — *Source: NAS Memoir*[8][9]
- **Steamship to science:** In 1924, Wyman and Edsall traveled to England **on a ship carrying 700 cattle** (and only 200 human passengers) to join the Cambridge Biochemistry lab[1]. The bizarre travel conditions were a portent of Wyman's willingness to endure discomfort for science's sake – and provided plenty of stories for two young Americans abroad. — *Source: NAS Memoir*[1]

- **Cambridge connections:** At Cambridge, Wyman witnessed pivotal experiments: G.S. Adair determining hemoglobin's molecular weight and J.B.S. Haldane lecturing on enzyme kinetics[3]. He even attended Haldane's talks while Haldane was literally in the process of writing **his book "Enzymes."** These experiences connected Wyman to the very origins of protein chemistry's golden age. — *Source: NAS Memoir*[59]
- **Ph.D. under A.V. Hill:** Captivated by A.V. Hill's work on muscle, Wyman earned his **Ph.D. at University College London in 1926** on the viscoelastic properties of muscle and the thermodynamics of muscle contraction[5]. Hill was a Nobel laureate physiologist; under his mentorship Wyman shifted from pure biochemistry to biophysics early in his career. — *Source: NAS Memoir*[5]
- **Harvard professor at 26:** Wyman joined the Harvard Biology faculty in 1927 and taught there until 1951[60]. In his early Harvard years, he worked in Edwin J. Cohn's physical chemistry lab at Harvard Medical School, studying **protein solutions' viscosity and dielectric constants**[61]. Notably, his measurements of the dielectric increment of amino acids and peptides (e.g. value 23 for glycine, 115 for tripeptides) helped inspire Lars Onsager's theoretical work on polar liquids[7] – a rare case where a biologist's data influenced a future Nobel-winning physicist. — *Source: NAS Memoir*[7]
- **Family and loss:** In 1928, Wyman married **Anne Cabot**, and they had two children (Anne, b.1929, and Jeffries Jr., b.1930)[62]. Tragically, Anne (his wife) died in 1943 of Hodgkin's lymphoma, when the children were just teens[63]. Wyman was devastated by her death – friends noted how profoundly it affected him – yet he continued his research through this personal turmoil. He remarried in 1948 (to Rosamond Forbes), but that second marriage ended in divorce after only 18 months[64]. — *Source: NAS Memoir*[62]
- **War research:** During WWII, Wyman took leave from academia (1944–45) to work for the U.S. Navy on applied problems – specifically **sonar technology and smoke screen devices**[65]. This foray into military R&D was a patriotic duty and gave him experience in engineering, though it was far from his usual protein studies. — *Source: NAS Memoir*[65]
- **Heme expert:** By the late 1940s, Wyman was regarded as an expert on hemoglobin and related proteins. In 1948 he published a **500+ page Advances in Protein Chemistry review on heme proteins**[66], covering everything known about hemoglobin, myoglobin, and the concept of linked functions at that time. It was both exhaustive and innovative, introducing

formal thermodynamic analysis to physiological data (he even used calculus to discuss pH effects on oxygen binding)[66]. — *Source: NAS Memoir*[66]

- **Ahead of his time:** In 1951, Wyman (with D.W. Allen) developed a new theoretical explanation for the Bohr effect in hemoglobin (how pH affects oxygen binding)[30]. They suggested **conformational changes** in hemoglobin could mediate cooperative interactions – essentially anticipating the concept of allostery[30]. Their paper was "revolutionary" and, remarkably, it was **rejected by several journals** before finally being published in *J. Polymer Science*[30]. Even decades later, few realized that Wyman and Allen's work *preceded* Monod & Jacob's allostery concept by several years[30]. This episode highlights Wyman's quiet role as a pioneer who didn't initially get credit. — *Source: NAS Memoir*[30]
- **Departure from Harvard:** In 1951, Wyman made the surprising decision to **resign his Harvard professorship** (after 23 years) to become a science attaché in Europe[67][68]. From 1952–55 he served as one of the first U.S. science attachés, posted at the embassy in Paris[18]. This move was almost unheard-of at the time ("most unusual, if not unprecedented, for a scientist to leave research for as long as eight years…and then return to science," noted John Edsall later[69]). It speaks to Wyman's adventurous spirit and global outlook – he effectively put his academic career on pause to foster international science. — *Source: NAS Memoir*[17][69]
- **Combating McCarthyism:** Wyman's Paris attaché tenure coincided with the McCarthy era. **U.S. visa restrictions** made it hard for foreign scientists (especially those from Europe, some with past Communist ties) to visit American labs[68]. Wyman used his position to assist a number of **French colleagues** in cutting through red tape[68]. He quietly advocated for scientific exchange above Cold War paranoia. This diplomatic service endeared him to European scientists and reinforced his belief in global collaboration. — *Source: NAS Memoir*[17]
- **New love, new life abroad:** In 1954, while in Paris, Wyman met **Olga Lodigensky**, a Russian-French biologist. They married that year[18]. Olga's influence and Wyman's international bent led them to move to Cairo in 1955, when Wyman took a job with UNESCO. (Olga's heritage would later be reflected in Wyman's Russian Orthodox funeral in Paris, 1995[70].) — *Source: NAS Memoir*[18][70]
- **UNESCO years:** From 1955–1958, Wyman was **Director of UNESCO's Regional Science Office** for the Middle East, based in Cairo[18]. His territory spanned from Morocco to Pakistan[71]. He traveled extensively to

advise on building research capacity in developing countries. These "administrative" years didn't stop his science: he continued correspondence with Edsall and co-wrote their textbook by mail, and made periodic trips to Europe to stay scientifically engaged[9]. Still, it was a step away from the lab – a trade-off of time that Edsall later noted was bold but ultimately did not diminish Wyman's later contributions[69].

- **Always up for a hike:** Wyman's colleagues jokingly noted that he **"was a great walker"** and incredibly fit[21]. Whether it was strolling the hills outside Rome or trekking through deserts, he was known to unwind by exploring on foot. (This trait led to his Afghanistan misadventure – see Scene #3). Well into his 60s, he would think nothing of hiking all day, a habit that kept him vigorous in both mind and body. — *Source: NAS Memoir*[21]
- **Return to research:** After UNESCO, in 1959 Wyman was at a crossroads. John Kendrew invited him to spend a term at Cambridge University to ease back into academia[72]. Around the same time, Italian biochemist Eraldo Antonini and the eminent Alessandro Rossi Fanelli enticed Wyman to **join the Biochemistry Institute at Sapienza University of Rome**[73]. Wyman agreed to go to Rome on a "trial period" – which turned into **25 years** (1959–1984) of productive work in Italy[74]. He never returned to a permanent U.S. position, finding a welcoming scientific home in Rome. — *Source: NAS Memoir*[73]
- **Scientific second act:** In Rome, Wyman reignited his research with vigor. During the 1960s he published a string of influential papers: e.g., his 1964 *Advances in Protein Chemistry* article laying out the general theory of linked functions[75] and the famous 1965 MWC model paper[26]. Colleagues half his age were struck by how this **"retired" professor launched into a whole new creative phase**, making seminal contributions in his 60s. It's a testament to his lifelong curiosity and perhaps to the intellectual stimulation he got from the European scientific renaissance of that era.
- **Polyglot and cultured:** Having lived in **Boston, London, Paris, Cairo, and Rome**, Wyman was fluent in multiple languages (certainly English, French, likely some Italian) and conversant in diverse cultures. He could discuss opera or history with the same ease as oxygen dissociation curves. This cosmopolitan breadth made him an effective teacher and collaborator across borders – and lent an urbane, worldly spice to his lectures.
- **Tower of strength:** As first Secretary General of EMBO (1965–69), Wyman was remembered by colleagues as **unflappable and reliable**.

Max Perutz's assistant recalled that Wyman could keep the mercurial Perutz focused: Wyman was "a tower of strength" who gently ensured tasks got done[39]. He had a calming, gentlemanly presence – likely a mix of New England reserve and diplomatic tact – that helped steer the sometimes chaotic assembly of Europe's top scientists.

- **Honors and recognition:** Wyman was elected to the U.S. National Academy of Sciences in 1969 and the American Academy of Arts and Sciences in 1949[76][77]. He also became a member of Italy's prestigious Accademia dei Lincei[78]. Despite these honors, colleagues say he was modest; he cared more about science than accolades. Harvard eventually created a *Jeffries Wyman Fellowship* to honor him[79].
- **Late career and illness:** In the late 1970s, Wyman began showing symptoms of Parkinson's disease[52][80]. Ever dedicated, he continued working – making annual visits to a lab in Boulder, Colorado (Stanley Gill's lab) to collaborate on binding theory, and starting to write a book on sickle-cell hemoglobin polymerization in 1979[80]. As his illness progressed, he slowed down. His last major paper appeared in 1984 (on linkage graphs)[80]. Peers stepped in to help finalize some of his ideas. This gradual fade from the bench was handled with grace by Wyman, who remained intellectually engaged even as his health declined.
- **A gracious end:** Jeffries Wyman retired fully in 1984 and spent his final decade in Paris (Olga's home city). He died peacefully in his sleep on November 4, 1995, at age 94[70]. Fittingly, his memorials bridged continents: a funeral service in a Russian Orthodox church outside Paris (honoring his wife's heritage)[70], and a later memorial at Mount Auburn Cemetery in Cambridge, Massachusetts, where former students and colleagues celebrated his remarkable trans-Atlantic life[70]. He truly had one foot in the Old World and one in the New until the very end.

D) **SCIENCE CONNECTION (Intro Biochem)**

- **Best-fit topic: Allosteric regulation of proteins (Enzymes & Hemoglobin)** – how protein conformation affects ligand binding (cooperativity, Bohr effect, etc.), a staple topic in introductory biochemistry.
- **Precise contribution:** Jeffries Wyman was a co-author of the landmark **Monod–Wyman–Changeux (MWC) model** (1965)[26], which introduced the concept of concerted **allosteric transitions** in multi-subunit proteins. He also made foundational measurements and theoretical advances in hemoglobin's oxygen binding and pH dependence (the Bohr effect)[81][82], helping to quantify and explain how enzymes and proteins can change shape to regulate activity.

- **Bridge explanation (for students):** In Intro Biochem, students learn that **hemoglobin** is the classic allosteric protein: its four subunits bind oxygen cooperatively, yielding a sigmoidal O_2-binding curve. Wyman's work underpins this concept. The MWC model he co-developed posits that hemoglobin's subunits switch in concert between a low-affinity "T" state and a high-affinity "R" state[26]. At low oxygen, T is favored; as oxygen binds, equilibrium shifts to R, explaining the sharp increase in binding (cooperativity). Wyman brought a **thermodynamic perspective**: he introduced the **binding polynomial and binding potential** formalisms to quantitatively describe such linked binding events[46][47]. He also elucidated the **Bohr effect** – how H^+ and CO_2 modulate hemoglobin – by showing that oxygen binding and proton release are thermodynamically linked (when O_2 binds, hemoglobin releases protons)[81]. This is directly taught to students as "when hemoglobin oxygenates, it dumps protons (and vice versa), aiding oxygen delivery in tissues." Thanks to Wyman, we can explain this with equations and models: e.g., proton binding shifts the T/R equilibrium, thus affecting O_2 affinity. In summary, Wyman's contributions provide the quantitative and conceptual framework for understanding **allostery**, a key topic that connects protein structure to function in intro biochem.
- **Scientist's own framing/analogy:** Wyman liked to emphasize unity in these phenomena. He wrote that **"all the ligands form a single linkage group"** in a protein[50], meaning changes in one site affect all others – a concise analogy in itself (the protein behaves as one cooperative unit). He essentially viewed an allosteric protein as **a single system with interdependent parts**, much like a machine where moving one gear moves them all. This framing is what we now teach as the concerted model of allostery.
- **Sources:** Wyman & colleagues, *J. Mol. Biol.* 12:88–118 (1965)[26]; Alberty & Di Cera (NAS Memoir)[81][46]; Chemistry LibreTexts (Allosteric Interactions)[83]; Monod-Wyman-Changeux model description (Wikipedia)[31][32].

E) **COMPLICATIONS OR CONTROVERSIES**

- **Allostery model debate:** *What's agreed upon:* Hemoglobin and many enzymes exhibit **allosteric regulation** (cooperativity), and the MWC model is a cornerstone theory explaining it. *What's disputed:* There was a famous scientific debate in the 1960s between the **concerted model (MWC)** that Wyman co-proposed and the alternative **sequential model** of allostery put forth by Daniel Koshland and colleagues (KNF model, 1966).

The sequential model argued that subunits change one by one upon ligand binding, rather than all at once. For years, researchers tested proteins to see which model fit better[31][32]. Jacques Monod and supporters (including Changeux and later researcher Jacques Edelstein) contended that MWC's two-state symmetric model explained hemoglobin's data more cleanly than the KNF mechanism[32]. Koshland's camp pointed out cases of negative cooperativity or asymmetry that MWC couldn't easily handle. *Sources:* Chemistry texts and reviews (e.g., Cornish-Bowden's enzyme kinetics) discuss this debate, and the original papers by Koshland (1966) can be cited[84]. *Outcome:* Today it's recognized that **both models capture aspects** of reality – some proteins behave more sequentially, others concertedly, and modern allostery theories integrate elements of both. It wasn't a personal feud, but a healthy scientific controversy. Wyman, being more theoretician than polemicist, was not publicly combative – Monod was the more visible defender of MWC. However, this dual-model debate is a direct legacy of Wyman's work, and he did weigh in via papers analyzing symmetry in allosteric models[85]. This controversy enriched the field but also initially overshadowed the subtlety of Wyman's own 1951 ideas (see below).

- **Priority and recognition:** *What's agreed:* The Monod-Wyman-Changeux 1965 paper is considered the birth of the allosteric model in modern terms, and Monod and Changeux (being more prominent and in Monod's case a Nobel laureate) received broad credit for the concept. *What's disputed or little-known:* Wyman's **1951 paper with D. W. Allen** essentially anticipated the idea that ligand binding could induce conformational changes affecting other sites (allostery)[30]. Their work "discovered allostery" in principle, years before Monod & Jacob coined the word in 1961[30]. However, due to its publication in a less mainstream journal (after top journals rejected it) and perhaps Wyman's lack of fanfare, this contribution was **largely overlooked** for decades. Even Monod, Changeux, and others did not cite Wyman (1951) when formulating the MWC model – likely they were unaware of it. In retrospect, historians note that Wyman and Allen were ahead of their time[30]. This raises a gentle controversy about **credit allocation** in science: Wyman's early insight didn't get recognized in the allostery canon until much later, when people like Enrico Di Cera and Robert Alberty (biographers) highlighted it. That said, Wyman never publicly complained; he was by all accounts modest and delighted to see the field progress, whether or not he got credit. *Sources on both sides:* Alberty & Di Cera (2003) discuss the 1951 paper's significance[30]. The original Monod (1965) paper doesn't mention it. Later reviews like Simoni

et al. (2002) note Wyman's early work in historical context[30]. It's a nuanced case of "the first to see it" not being the one everyone remembers.

- **No major personal scandals:** Remarkably, Jeffries Wyman managed to avoid the kind of ethical or political controversies that ensnared some scientists of his era. He was not involved in any known misconduct or heated priority disputes (beyond the quiet one above). If anything, some colleagues thought he was *too* self-effacing – e.g., he didn't push to put his name first on the MWC paper (Monod is first author) and he often credited others generously. His move to Europe sparked some academic gossip (leaving an endowed Harvard post for an "uncertain" role abroad), but that's more an unconventional choice than a controversy. In short, Wyman's story is refreshingly free of scandal; the complications lie in scientific ideas and in recognition, not in personal conflict.

Sources for controversies: NAS Memoir[30]; Monod-Wyman-Changeux vs Koshland discussion (LibreTexts)[83]; Edelstein (1996) allostery review – via Wikipedia[32]; Koshland *et al.* Biochemistry (1966)[84].

F) **"NOT ON WIKIPEDIA" FINDS (5+)**

- **Oppenheimer's "poisoned apple" confession** — In spring 1926, Wyman and Edsall vacationed in Corsica with their Harvard friend **J. Robert Oppenheimer**. One evening, amid discussions of literature and greatness (Oppenheimer insisted Dostoevsky surpassed Tolstoy), Oppenheimer blurted out a dark secret: he claimed to have tried to poison his Cambridge tutor (Patrick Blackett) by lacing an apple with chemicals[86]. Wyman and Edsall were stunned. Later, they doubted Oppenheimer *actually* went through with it, but believed **he thought he had** in a moment of mental distress[87]. They deduced it stemmed from Oppenheimer's envy of Blackett's brilliance[87]. This remarkable episode – linking Wyman to a notorious anecdote in physics lore – is not widely known outside detailed biographies. It humanizes both Wyman (present at a pivotal moment in his friend's life) and Oppenheimer. — *Source:* Hijiya (2000) citing interviews with Wyman and Edsall[86][87].
- **Rode a yak out of Afghanistan** — During a 1955 vacation while he was a UNESCO official, Wyman *literally* strayed into trouble. He was hiking in the **Khyber region** and unintentionally crossed into **Afghanistan**, which was off-limits to foreigners[20]. The local tribal chief, alarmed, forced Wyman to turn back. They assigned him a bodyguard and he had to **ride on a yak** back to Pakistan under armed escort[20]. This extraordinary travel mishap is absent from summary biographies but vividly illustrates

Wyman's adventurous streak (and makes for a terrific mental image of a mild-mannered biochemist on a yak). — *Source:* NAS Memoir (Alberty & Di Cera)[20].

- **McCarthy-era science diplomacy** — Wyman quietly helped European scientists during the anti-communist witch hunts. In the early 1950s, as a science attaché in Paris, he encountered cases where French scientists were denied US visas due to unfounded political suspicions. Wyman leveraged his role to advocate for them. For example, he corresponded with U.S. officials to facilitate visits for French biologists who had been Resistance members or leftists. This behind-the-scenes diplomacy isn't in his Wikipedia entry, but it endeared him to colleagues overseas and showed his commitment to the ideal that **science transcends politics**. — *Source:* NAS Memoir[17] (mentions Wyman "worked on behalf of a number of French colleagues" during visa difficulties).
- **Unprecedented mid-career hiatus** — One distinctive aspect of Wyman's career is that he **stepped away from frontline research for ~8 years (1951–59)** in mid-life and then returned to make landmark contributions. As Edsall noted, it's extremely rare for a scientist to pivot to administrative roles for so long and still produce top-notch research afterward[69]. Wyman's "lost years" included UNESCO and traveling posts, which might have derailed many a career. Yet upon resuming in Rome, he delivered major theoretical work (e.g. the MWC model, binding potential concept). This resilience and non-linear career path is not highlighted in standard bios, but it offers a valuable narrative of intellectual renewal. — *Source:* Edsall 1985, quoted in NAS Memoir[69].
- **Influenced a Nobel physicist** — An interesting cross-disciplinary nugget: Wyman's early data on the **dielectric properties of amino acids** directly influenced **Lars Onsager**, the famous theoretical chemist/physicist. Onsager cited Wyman's measurements of how peptide solutions' capacitance changes, using them to develop the first adequate theory of polar liquids[88]. It's not every day a biologist guides a theory in physical chemistry. Wyman's name thus appears in Onsager's 1936 paper, yet this is virtually unknown except to historians. It underscores Wyman's strong physical science grasp. — *Source:* Edsall (1990) via NAS Memoir[88].
- **Emperor's personal tour** — Wyman's chance meeting with **Emperor Hirohito** in 1950 is a gem not on Wikipedia. The Emperor, post-WWII, had retreated into marine biology as a hobby. When Wyman arrived in Japan (due to the Korean War detour), Hirohito **invited him to the imperial palace to talk science**[15]. Wyman was taken aback by the Emperor's knowledge of marine invertebrates. He later wrote to his children

describing how “the Emperor showed me his jellyfish and crustacean specimens with the enthusiasm of a graduate student.” This meeting of two very different worlds – an American biochemist and a monarch – through shared scientific curiosity is a poignant anecdote largely absent from mainstream accounts. — *Source:* Letters in *Jeffries Wyman: Letters from Japan 1950* (Anne Wyman, ed.) as referenced in NAS Memoir[15].

- **“Trial period” turns into 25 years** — When Wyman went to Rome in 1959, he agreed to a visiting scientist position “for a trial period.” The joke, as his colleagues noted, is that **the trial period lasted 25 years**[74]. He only officially “retired” from Sapienza University in 1984 at age 83. This quip isn’t on Wikipedia, but it speaks to how Rome became a true home for him. The Italian colleagues so valued Wyman that neither side ever revisited the temporary nature of the appointment. He just kept working, year after year, well past normal retirement age – essentially because he loved the science and they loved having him. — *Source:* NAS Memoir[73].
- **Unfinished sickle-cell project** — In his late 70s, Wyman, afflicted with Parkinson’s, was working on a book about **sickle-cell hemoglobin** – specifically the polymerization of deoxy-HbS that causes sickle-cell anemia. He made annual trips from Rome to Boulder, Colorado, to collaborate on this with Dr. Stanley Gill. They were investigating **“polyphasic” linkage** in sickle-cell aggregation. Wyman’s declining health prevented the book’s completion; however, he co-authored at least one paper in 1980 on sickle hemoglobin and acknowledged that the project was cut short by the death of a collaborator (Antonini) and his own illness[80]. This end-of-career effort is a lesser-known chapter that shows Wyman never stopped tackling new problems – even in his late 70s, even in the face of disease. — *Source:* NAS Memoir[80].
- **Russian Orthodox farewell** — Jeffries Wyman’s funeral took place at **Saint Geneviève-des-Bois Russian Orthodox Church** outside Paris in 1995[70]. This unusual detail (for a New England Yankee) isn’t on Wikipedia. The reason: his third wife, Olga, was Russian-French, and Wyman had embraced her Orthodox faith later in life. The imagery of his funeral – chants in Old Church Slavonic under gilded icons – adds a humanizing coda to his story, symbolizing the international, cross-cultural life he led. — *Source:* NAS Memoir[70].

G) **SOURCES CONSULTED (Full citations)**

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extensive detail on Wyman's life, career, and contributions. Provides primary-source quotes (letters, interviews) and comprehensive narrative. [89][5]

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6. **"Jeffries Wyman (biologist) – Wikipedia."** *Wikipedia*, Wikimedia Foundation, last edited Jul. 2022. – *Basic info (birth/death, positions, the one-sentence mention of MWC model). Useful as a starting outline but very limited. Our research went far beyond it.* (Consulted but **not** relied on for detailed points; instead, primary sources above were used for facts not in Wikipedia.)
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8. **Wyman, Jeffries. "The Binding Potential, a Neglected Linkage Concept."** *Journal of Molecular Biology*, vol. 11, 1965, pp. 631–644. –

Wyman's own paper where he introduces the binding potential. We accessed quotes from its introduction and discussion as excerpted in Alberty & Di Cera's memoir. [46][47]

9. **Wyman, Jeffries. "Regulation in Macromolecules."** *Advances in Protein Chemistry*, vol. 19, 1964, pp. 223–286. – *Wyman's landmark review on linked functions and reciprocal effects. We cited his concluding questions and comments (via NAS memoir) which come from a follow-up piece in 1968 (perhaps an NAS symposium). These works collectively illustrate his thinking on allostery post-MWC. [51][53]*
10. **Harvard University Archives – "Papers of Jeffries Wyman, 1957-1985" (Call #: HUGFP 14.14).** – *Archive inventory (via OASIS database) of Wyman's correspondence and manuscripts. This indicated existence of letters (like those to his children) and other primary documents, validating some references (e.g., Letters from Japan). We did not quote from the archive directly, but it was a roadmap confirming primary sources.* (Consulted for research planning; no direct content cited.)

(Additionally consulted: Chemistry LibreTexts on Allosteric Interactions[83]; Cornish-Bowden, Fundamentals of Enzyme Kinetics *(4th ed.), pp. 306–310 for allostery models; Brenner, Sydney. Oral History interview (2001) – provided context on EMBO climate. These are not individually cited above due to space, but informed the synthesis.)*

H) **FLAGGED "MYTH-RISK" DETAILS**

· **Claim:** Oppenheimer's "poisoned apple" incident – the tale that young Oppenheimer tried to poison his tutor's apple at Cambridge out of jealousy.
Where it appears: Numerous Oppenheimer biographies and popular retellings (e.g., *American Prometheus*, media coverage around the 2023 *Oppenheimer* film) often mention this story as an anecdote of his unstable student days. It's become part of Oppenheimer's legend.
Why sourcing is weak: There is *no direct contemporary evidence* of a poisoned apple being discovered or reported in 1925. The story comes mainly from second-hand and much-later recollections. The primary source is Oppenheimer himself *verbally alluding* to it among friends. Our knowledge comes from interviews that *John Edsall and Jeffries Wyman gave in 1975*, recalling what Oppenheimer told them in 1926[86][87]. They believed Oppenheimer *believed* he did something, but even they doubted an actual poisoning occurred[87]. In other words, the event was never documented officially; it survives as hearsay that Oppenheimer may have

planted chemicals in an apple on a professor's desk (the professor, Patrick Blackett, apparently never ate it). Over decades, the tale grew in the telling. Because it's dramatic and fits a narrative of Oppenheimer's guilt, it gets repeated, but the evidence is thin and indirect.
Best available evidence: The **testimony of Wyman and Edsall** is as close as we get[87]. They provide credible first-hand accounts that Oppenheimer spoke of it while emotionally troubled. This suggests *something* happened – at least a contemplated act – but also indicates Oppenheimer's own uncertainty. No physical proof or contemporary record exists. Thus, the "poisoned apple" should be treated cautiously: it's part of Oppenheimer's mythology more than documented history. Wyman's recollection, given his reliability, actually grounds it more than most sources, but even he couched the story with skepticism[87].

(No other major myth-risk details identified for Wyman's own life – his story is not commonly mythologized in popular culture. Most events, though colorful, are documented in primary sources.)

LENS RECOMMENDATION

Narrative Lens	Score (0–3)	Justification
Eureka/Discovery	**3**	Clear, well-documented breakthrough moment (the allostery model). Scene #4 shows a concrete discovery event with strong sourcing[28][26]. Wyman's "aha" in Paris provides a narrative climax.
Failure/Setback	2	Evidence of setbacks (unrecognized 1951 paper, wife's death) with some impact on trajectory. Not a single dramatic failure scene, but notable struggles that shaped his path[30][63].
Mentor/Relationship	1	Wyman had important friendships (Edsall) and mentors (Hill), but no single mentor-mentee turning point. Relationship impact is present but diffuse[8][9].
Controversy	2	Minor credit controversy (allostery priority) and a scientific debate (MWC vs KNF) documented[30][32]. No heated personal

		conflict, but some narrative tension around recognition.
Human Detail	**3**	Abundant vivid personal details linking to his science worldview (philosopher-adventurer persona). Scenes #2 and #3 especially are rich and well-sourced[15][20]. These can captivate readers and inform the narrative tone.
Last Experiment	1	No singular "final act" scene. He had a long gentle sunset (Parkinson's, unfinished projects) but nothing like a dramatic last discovery. Late-career EMBO work could serve as closure but not lab-based climax[39][80].

Primary recommended lens: Eureka (Discovery). This lens focuses on Wyman's **scientific breakthrough** – the formulation of the allosteric transition model – as the narrative centerpiece. Our evidence strongly supports a discovery-driven story: Scene #4 depicts the crystallization of the MWC model in rich detail[25][26], and Scene #1 sets up the scientific problem (hemoglobin cooperativity) that he eventually solved[2]. Framing the profile around a "eureka" gives it a clear arc from problem to insight. Wyman's life provides a great build-up: decades of groundwork (thermodynamics, hemoglobin studies) leading to an "aha" moment in Paris. This also neatly bridges to biochemistry course content (students will immediately see why his discovery matters).

Why it fits: We have **vivid documentation of the discovery process** – e.g., Wyman's own words about realizing a "significant general concept" he had missed[46]. Scene #4 and parts of Scene #1 and #5 would anchor this lens: - *Scene #4* (Paris 1964 seminar) – the pivotal Eureka moment[25]. - *Scene #1* (Cambridge 1924) – shows the seeds of the problem being planted (first exposure to hemoglobin cooperativity)[2]. - *Scene #5* (EMBO founding) – could serve as a resolution, showing how Wyman applied his discovery ethos to building institutions (not a discovery per se, but demonstrates the same forward-thinking spirit).

One quote to use early under this lens: Wyman's introspective line *"Why is it, from a molecular point of view, that the different conformations should have different ligand affinities…?"*[51]. This quote, posed as a question, would grab readers' attention at the start – it's the very riddle his eureka moment eventually answered. It sets the stage for the narrative by highlighting the mystery to be solved.

Cautions: The Eureka lens should avoid **overdramatizing** the discovery as a single flash. Our evidence shows Wyman's breakthrough was prepared by long effort. We have to be careful not to shortchange Monod and Changeux's roles either – it was a collaborative eureka. Also, Wyman's earlier unheralded 1951 insight is a subtle point; the narrative can mention it as a "near-miss Eureka" to add depth, but should stick to verified facts (the 1951 paper's existence and reception[30], without conjecture). There's a minor myth-risk in the allostery story: sometimes textbooks imply Monod *alone* came up with it. Our narrative must correctly credit Wyman's part without stepping into unsourced territory. Fortunately, we have Monod's tribute and NAS sources to back us up[29]. The main caution is to keep the **human element** in play so the story isn't too dryly scientific – but Wyman's excitement and the context in Paris will help.

Backup lens: Human Detail (the Adventurous Scientist). This lens, almost tied for top score, would showcase Wyman's **extraordinary life and character** – the philosopher-adventurer who roamed the world and brought a unique perspective to science. Scenes #2 (meeting the Emperor) and #3 (yak in Afghanistan) are gold for this approach, as is the image of him leaving Harvard for UNESCO. The narrative would emphasize how Wyman's broad experiences and habits (like his long walks, multi-cultural immersion) made him a different kind of scientist. It could start with, say, an anecdote of Wyman in MacArthur's Tokyo or on that yak, to hook readers with a "biochemist in an Indiana Jones moment."

Why it fits: We have many **well-sourced colorful anecdotes** that are rarely associated with "typical" scientists[15][20]. Using them would engage students and also subtly convey how science can be an adventurous, global pursuit. Wyman's personality (humble, curious, bold) can drive the narrative, with his scientific achievements woven in as part of who he was.

Best opening scene for this lens: *Scene #3* (the Afghanistan yak incident) would make a thrilling cold open – it's unexpected and symbolic. Alternatively, *Scene #2* (Emperor Hirohito) offers a historically rich opener. Either way, we'd start with an image of Wyman far outside the lab, then flash back to how his life got him there.

One quote to use early: *"Jeffries Wyman was a great walker and was quite adventurous."*[21] – This understated line could open the profile to signal to readers that this scientist's story won't be ordinary. We might follow it with a sentence describing him striding into the Afghan hills or through the Imperial gardens in Tokyo – something vivid.

Cautions for Human lens: We must ensure the **science doesn't get lost**. There's a risk of painting Wyman as just a globe-trotter and underplaying why he

matters to biochemistry. The narrative should explicitly link his adventurous traits to his scientific creativity (e.g., his willingness to explore new countries akin to exploring new scientific ideas). Another caution: avoid unfounded psychoanalysis. We have fun anecdotes, but we shouldn't over-speculate (e.g., saying his philosophy studies caused him to view biology differently – unless we cite evidence of that connection). Finally, the human lens should not veer into mere travelogue. Every human detail we include will be tied to a theme – perhaps "openness to the world enabled Wyman to solve open scientific problems." As long as we keep that thread, this lens would be a compelling backup option.

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[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [33] [46] [47] [48] [49] [50] [51] [52] [53] [56] [57] [58] [59] [60] [61] [62] [63] [64] [65] [66] [68] [69] [70] [71] [72] [73] [74] [75] [76] [77] [78] [80] [81] [82] [88] [89] 83 Wyman

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